ViT-RealWaste: A Leakage-Aware Benchmark of a Vision Transformer against CNNs for Landfill Waste Classification

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Abstract. Automated sorting of landfill waste needs classifiers that stay reliable on cluttered, deformed and contaminated items. The RealWaste dataset of Single, Iranmanesh and Raad showed that the training domain dominates performance: CNNs trained on pristine images reach at most 49.69% accuracy on real landfill images, against 89.19% when trained on RealWaste itself. All architectures compared there were convolutional. We benchmark a Vision Transformer (ViT-B/16) against VGG-16, MobileNetV2 and Inception V3 on the public RealWaste release (4,752 images, nine classes) under one leakage-aware protocol: stratified splitting *before* any augmentation, class-weighted loss, AdamW, plateau-driven learning-rate decay and three-stage progressive fine-tuning. ViT-B/16 attains 93.35% accuracy and the highest macro-F1 (93.86%); Inception V3 ties in accuracy (93.35%, macro-F1 93.59%), while VGG-16 and MobileNetV2 trail by 2.5 and 4.8 points. Bootstrap 95% confidence intervals (about ± 2.2 points on 481 test images) overlap for the two leading models, so their ranking is not statistically resolved. ViT-B/16 recovers the hardest class, Miscellaneous Trash, best (84% recall versus 72–78%), but needs $2.2\times$ the training time and $3.9\times$ the parameters of Inception V3. The same confusions (Miscellaneous Trash; Metal–Plastic) recur in every architecture, which points to label ambiguity rather than model family as the main remaining bottleneck.

Keywords: Vision Transformer, ViT-B/16, RealWaste, landfill waste classification, transfer learning, progressive fine-tuning, bootstrap confidence intervals

1. Introduction

Waste disposal is a measurable contributor to climate forcing: Single et al. report that the waste sector accounted for 3.48% of global greenhouse-gas production in 2020 and that landfill gas contributed about 20% of worldwide methane emissions in the following year [1]. Because gas generation depends strongly on the organic fraction of the waste, auditing the *composition* of what arrives at a landfill, at the level of food, vegetation, metals, plastics and so on, is a prerequisite for credible landfill modelling. Manual audits are subjective and labour-intensive, which motivates image-based classification.

The RealWaste study [1, 2] addressed a weakness of earlier work: most waste datasets show pristine, isolated objects, whereas items reaching a landfill are commingled, deformed and contaminated by decomposing organics. Training five CNNs on a pristine composite dataset (DiversionNet, built from TrashNet and other open data) gave only 26.82–49.69% accuracy on real landfill images; training the same networks on RealWaste raised this to 85.65–89.19% (Fig. 1b). The conclusion was that sampling training data from the real environment is essential, and that Miscellaneous Trash and visually similar material pairs remain difficult.

Every network in that study was convolutional. Vision Transformers (ViT) [3] replace local convolution with global self-attention over image patches, so each token can interact with every other token from the first layer onwards [4], whereas the effective receptive field of a CNN grows slowly with depth [5]. Whether this property helps on cluttered landfill imagery is an empirical question. Follow-up CNN studies on RealWaste exist [6], and transformers have been applied to other waste tasks [7, 8], but we are not aware of a published evaluation of ViT-B/16 on RealWaste (literature search of October 2026; unpublished work cannot be excluded).

We therefore ask a narrow question and answer it carefully: *under an identical, leakage-aware protocol, does ViT-B/16 outperform representative CNNs on RealWaste, and by how much relative to sampling noise?* Our contributions are:

1. A fully specified, reproducible benchmark of ViT-B/16, VGG-16, MobileNetV2 and Inception V3 on the public RealWaste release, using a split-before-augment protocol (Sec. 4).
2. An uncertainty analysis: 95% bootstrap intervals reconstructed from the confusion matrices show that the two best models are statistically indistinguishable on this test set (Sec. 5).
3. A diagnostic analysis of error structure, computational cost and the contribution of each fine-tuning stage, which shows that the dominant failure modes are shared by all architectures (Secs. 5–6).

We do not claim a new state of the art: other authors report higher accuracies on RealWaste under different, partly non-comparable protocols (Sec. 2).

2. Related Work

Waste classification with CNNs. TrashNet [9] (six classes, controlled backgrounds) became the standard benchmark, with many CNN studies built on it [10, 11, 12]. TACO [13] and the detection work of Majchrowska et al. [14] moved toward in-the-wild imagery. RealWaste [1] contributed an operational-landfill dataset with nine material classes and quantified the cost of training on pristine data. Younis and Obaid [6] later compared five CNNs on RealWaste and report 96.31% for EfficientNet. They state that the data were enlarged to 19,008 images ($4 \times 4{,}752$) and split 50/20/30; because the text does not make the order of augmentation and splitting unambiguous, we cannot verify that augmented siblings of an image never cross the train/test boundary, and we do not use that figure as a comparison point (see Sec. 4.2).

Transformers. ViT [3] builds on the Transformer [15];

DeiT [16] and Swin [17] reduced its data and compute requirements. In waste applications, Tanwer et al. [7] fine-tune ViT, Swin and PVT for garbage-bin fullness detection, and Zhou et al. [8] propose a hybrid CNN–Transformer network evaluated on TrashNet and other sets. Studies on ImageNet-scale data suggest that ViTs rely more on shape than CNNs do [18, 19]; this motivates, but does not prove, an advantage on landfill imagery.

3. The RealWaste Dataset

3.1 Provenance and the original work

RealWaste was created by Sam Single, Saeid Iranmanesh and Raad Raad at the School of Electrical, Computer and Telecommunications Engineering, University of Wollongong, Australia [1]. The images were captured during the biannual residential waste audit at the Whyte's Gully Waste and Resource Recovery Centre in Wollongong (New South Wales), from municipal, recycling and organic bin streams, at a resolution of 524×524 pixels. Single et al. organised the content into three primary material groups (recyclable inorganics, divertible organics and inorganic waste requiring landfill) refined into nine labels. In the same study they built DiversionNet, a pristine counterpart, and trained VGG-16, DenseNet121, Inception V3, InceptionResNetV2 and MobileNetV2 on both datasets, which established the central result that training on real landfill material is required (Fig. 1b). Their error analysis identified Miscellaneous Trash (74.00% class accuracy for their best model), cardboard confused with printed metal/plastic/paper items, transparent plastic versus glass, and metal versus plastic as the weak points [1]. We revisit these findings with a different architecture family in Sec. 5.

The dataset is publicly available from the authors' repository [2], the UCI Machine Learning Repository and IEEE DataPort. The public release contains 4,752 images; the preprint describes 4,808 captured samples (4,327 for training and 481 held out) [1], so we use the public release and our own split. The UCI page lists the licence as CC BY 4.0, whereas the dataset reference in [1] and IEEE DataPort list CC BY-NC-SA 4.0; we follow the more restrictive terms, which cover our non-commercial research use.

3.2 Classes, imbalance and partitioning

Table 1 lists the nine classes. The imbalance is moderate (largest/smallest class ratio $921/318 \approx 2.9$), with Plastic and Metal dominating and Textile Trash the minority. Miscellaneous Trash is by definition a residual class and is therefore visually heterogeneous. We split each class independently into 80% training, 10% validation and 10% test images (flooring the first two counts and assigning the remainder to test; random seed 42), giving 3,798, 473 and 481 original images. The test set is never used for training, augmentation, model selection or learning-rate control.

3.3 Caveats

RealWaste provides no group identifiers, so photographs of the same physical item, or of items from the same bin, may fall into different partitions of a random split. This can make all absolute scores optimistic; it applies equally to every model compared here and to published random-split results. In addition, the class definitions overlap by construction (transparent plastic and glass; printed cardboard and paper; Miscellaneous Trash versus everything), so a part of the remaining error may be irreducible at the label level.

Table 1. RealWaste classes in the public release and our stratified split (original images; descriptions summarised from [1], examples illustrative).

Class	Typical content	All	Train	Val	Test
Cardboard	boxes, printed packaging	461	368	46	47
Food Organics	discarded food, e.g. bread	411	328	41	42
Glass	bottles, jars, bottle tops	420	336	42	42
Metal	cans, wiring, caps	790	632	79	79
Misc. Trash	residual, fits no other class	495	396	49	50
Paper	paper, paper packaging	500	400	50	50
Plastic	bottles, containers, film	921	736	92	93
Textile Trash	fabric, clothing	318	254	31	33
Vegetation	leaves, plant matter	436	348	43	45
Total		**4,752**	**3,798**	**473**	**481**

4. Method

Fig. 2 summarises the seven-phase protocol shared by all four backbones; only the input size, batch size and unfreezing indices differ.

4.1 Task and metrics

Given an image $\mathbf{x}$, a classifier maps it to class probabilities over $K = 9$ labels. With per-class precision P_k, recall R_k and support n_k among N images,

$$\mathrm{F1}_k = \frac{2P_kR_k}{P_k+R_k}, \quad \mathrm{F1}_{\mathrm{macro}} = \frac{1}{K}\sum_{k=1}^{K}\mathrm{F1}_k, \quad \mathrm{F1}_{\mathrm{w}} = \sum_{k=1}^{K}\frac{n_k}{N}\mathrm{F1}_k. \tag{1}$$

Weighted F1 drives model selection on the validation set; we report accuracy and macro-F1 on the test set, because macro-F1 gives each class equal weight and is therefore sensitive to the minority classes.

4.2 Leakage-aware preprocessing and augmentation

Why order matters. Suppose every original image is expanded into k augmented copies and the *pooled* images are then split at random with test fraction ρ. A test copy then has k siblings, each in the training set with probability $1-\rho$, so the probability that at least one sibling is in training is approximately

$$p_{\mathrm{leak}} = 1-\rho^k. \tag{2}$$

For $\rho = 0.3$ and $k = 3$ this is 97.3%: the model would be tested on near-duplicates of training images. We avoid the problem by construction: the split (P1) precedes augmentation, and augmented images are generated only from images of the same partition.

Offline stage. After splitting, a geometric pipeline built with Augmentor [20] generates extra images for the training

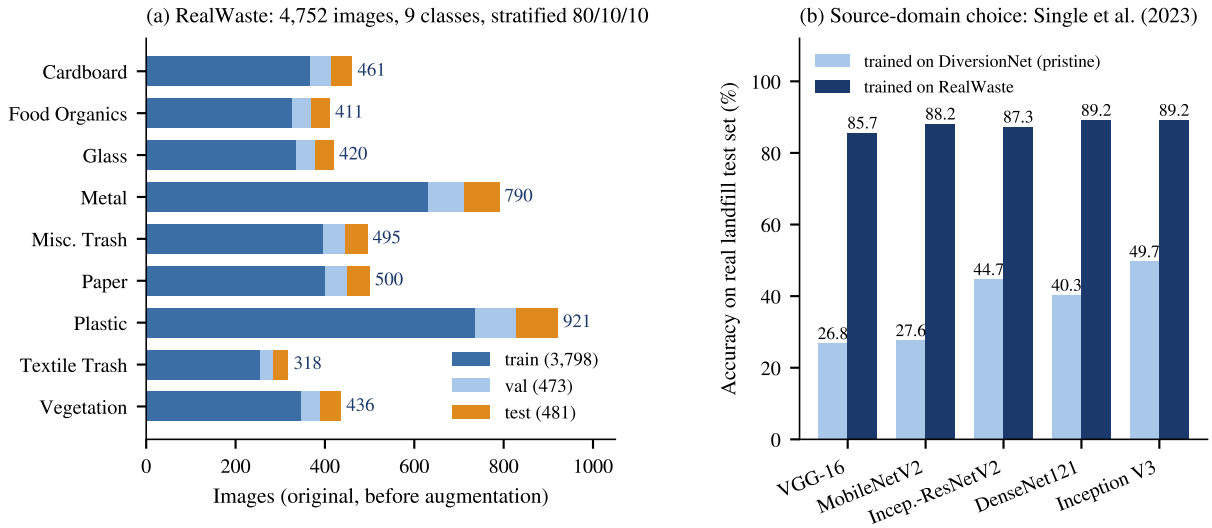


Figure 1. (a) Class distribution and our split of the public RealWaste release. (b) Source-domain effect reported by Single et al. [1]: accuracy on the real landfill test set for networks trained on pristine DiversionNet versus RealWaste.

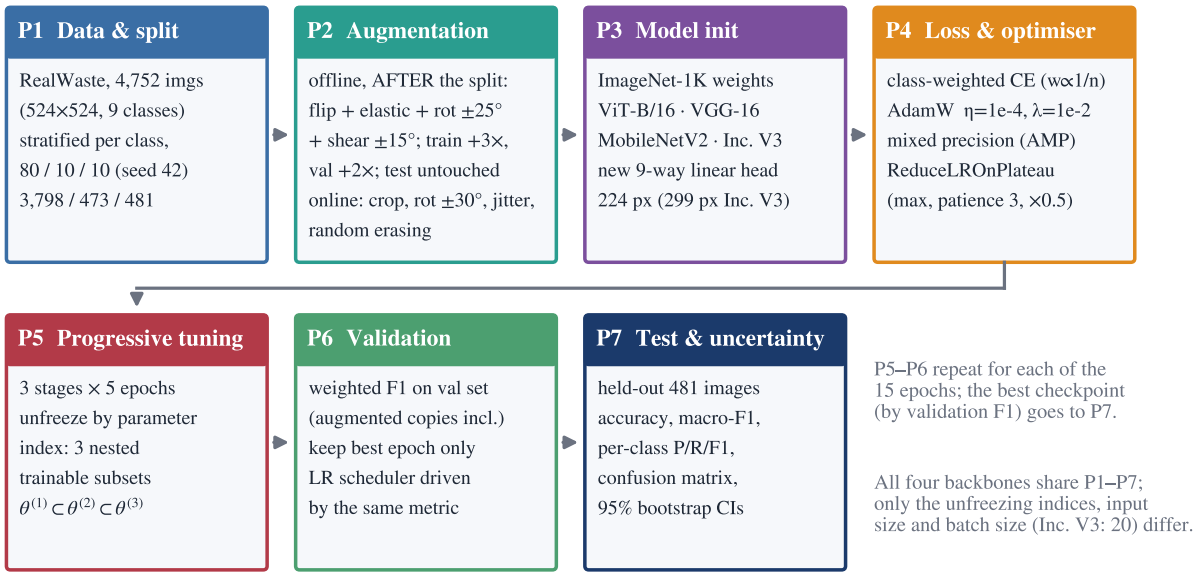


Figure 2. Seven-phase protocol. Splitting precedes augmentation (P1→P2), so augmented siblings never cross partition boundaries; P5–P6 repeat for 15 epochs and the best validation checkpoint is evaluated once on the untouched test set (P7).

set (3× its size) and the validation set (2×). Every generated image passes through horizontal flip, random elastic distortion, rotation within ±25° and shear within ±15°. Together with the originals, this yields 15,192 training and 1,419 validation images; the test set is not augmented. (These counts are derived from the multipliers, as the generated files are stored alongside the originals.) Geometric transformations were reported to help more than photometric ones in several studies cited by Single et al., whose own pipeline likewise combined flips, elastic distortion, rotation and shear [1].

Online stage. At each iteration the training images additionally undergo a random resized crop (area scale 0.7–1.0, output 224^2, or 299^2 for Inception V3), horizontal flip ($p = 0.5$), rotation within ±30°, colour jitter (brightness, contrast, saturation 0.3; hue 0.15), and random erasing ($p = 0.3$). Evaluation uses a deterministic resize to 256 px (320 px for Inception V3) followed by a centre crop. All images are standardised with the ImageNet statistics used in backbone pre-training,

$$\hat{x}_c = \frac{x_c - \mu_c}{\sigma_c}, \quad \boldsymbol{\mu} = (0.485, 0.456, 0.406),\ \boldsymbol{\sigma} = (0.229, 0.224, 0.225). \tag{3}$$

Relative to the original study, which augmented offline only and trained at 524 × 524, the working resolution is much lower, which matches the backbone pre-training resolution and reduces

Table 2. Backbones. Parameters include the 9-way head (Inception V3 without its unused auxiliary branch); GFLOPs are the torchvision model-zoo values for 1000-class heads. S1/S2 give the trainable portion at stages 1 and 2.

Model	Params	GFLOPs	Input	Trainable at S1 / S2
MobileNetV2	2.2 M	0.3	224^2	last block, final conv, head / from block 8
VGG-16	134.3 M	15.5	224^2	99.1% / 99.9%
Inception V3	21.8 M	5.7	299^2	Mixed_7b/7c, head / from Mixed_6c
ViT-B/16	85.8 M	17.6	224^2	96.4% / 99.1%

memory.

4.3 Models

ViT-B/16 [3] (Fig. 3). An image $\mathbf{X} \in \mathbb{R}^{224\times224\times3}$ is cut into $N = HW/P^2 = 196$ patches of $P = 16$ pixels. Each flattened patch $\mathbf{x}_p^i \in \mathbb{R}^{768}$ is projected by $\mathbf{E}$; a learnable class token $\mathbf{x}_{\text{cls}}$ and position embeddings $\mathbf{E}_{\text{pos}}$ are added:

$$\mathbf{Z}^{(0)} = \left[\mathbf{x}_{\text{cls}}; \mathbf{x}_p^1\mathbf{E}; \ldots; \mathbf{x}_p^N\mathbf{E}\right] + \mathbf{E}_{\text{pos}} \in \mathbb{R}^{197\times768}. \tag{4}$$

Each of $L = 12$ encoder layers applies layer normalisation [21] before multi-head self-attention (MHSA) and before an MLP, both wrapped in residual connections (pre-LN, which stabilises optimisation [22]):

$$\mathbf{Z}'^{(\ell)} = \text{MHSA}\big(\text{LN}(\mathbf{Z}^{(\ell-1)})\big) + \mathbf{Z}^{(\ell-1)}, \tag{5}$$

$$\mathbf{Z}^{(\ell)} = \text{MLP}\big(\text{LN}(\mathbf{Z}'^{(\ell)})\big) + \mathbf{Z}'^{(\ell)}. \tag{6}$$

With $h = 12$ heads of width $d_k = 64$ and $\tilde{\mathbf{Z}} = \text{LN}(\mathbf{Z}^{(\ell-1)})$,

$$\text{head}_i = \text{softmax}\Big(\frac{\tilde{\mathbf{Z}}\mathbf{W}_i^Q(\tilde{\mathbf{Z}}\mathbf{W}_i^K)^\top}{\sqrt{d_k}}\Big)\tilde{\mathbf{Z}}\mathbf{W}_i^V, \tag{7}$$

$$\text{MHSA}(\tilde{\mathbf{Z}}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_{12})\,\mathbf{W}^O. \tag{8}$$

The $1/\sqrt{d_k}$ factor keeps the logit variance near one: for unit-variance queries and keys, $\text{Var}(\mathbf{q}\cdot\mathbf{k}) = d_k$, and without scaling the softmax saturates and its gradients vanish. The MLP is $\text{MLP}(\mathbf{z}) = \text{GELU}(\mathbf{z}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2$ with hidden width 3072 [23]. The attention matrix of each head is 197×197, so each layer costs $O(N^2D)$ in the sequence length, and every patch interacts directly with every other patch in every layer. Finally the CLS output is classified, $\hat{\mathbf{y}} = \text{softmax}\left(\text{LN}(\mathbf{z}_L^{[0]})\mathbf{W}_{\text{head}} + \mathbf{b}\right)$ with $\mathbf{W}_{\text{head}} \in \mathbb{R}^{768\times9}$.

CNN baselines. VGG-16 [24] is a plain stack of 3×3 convolutions with a large fully connected head; MobileNetV2 [25] uses inverted residual blocks with depthwise convolutions; Inception V3 [26] uses parallel factorised convolution branches at 299^2 input. All four networks start from ImageNet-1K weights [27, 28] and receive a new 9-way linear head. Table 2 gives their size and cost.

4.4 Training protocol

Class-weighted loss. With class weights $w_k = 1/n_k$ computed on the training set and batch labels y_i, the loss is the weight-normalised cross-entropy

$$\mathscr{L} = \frac{\sum_i w_{y_i}\ell_i}{\sum_i w_{y_i}}, \qquad \ell_i = -\log\hat{y}_{i,y_i}. \tag{9}$$

The normalisation makes $\mathscr{L}$ invariant to the scale of w, and since $\pi_k w_k$ is constant for class priors π_k, every class contributes equally to the expected gradient regardless of its frequency; this targets the Plastic/Textile imbalance without resampling (cf. [29]).

Optimiser. We use AdamW [30] with learning rate $\eta = 10^{-4}$ and weight decay $\lambda = 10^{-2}$. With $\hat{\mathbf{m}}_t, \hat{\mathbf{v}}_t$ the bias-corrected first and second gradient moments, the update decouples decay from the adaptive step:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\,\frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t} + \varepsilon} - \eta\lambda\,\boldsymbol{\theta}_t. \tag{10}$$

Forward passes use mixed precision (fp16 autocast with loss scaling) [31].

Plateau-driven decay. After each epoch e the validation weighted F1, F_e, is passed to a reduce-on-plateau rule (patience $p = 3$, factor $\gamma = 0.5$): if F_e has not exceeded its running best for $p+1$ consecutive epochs, then $\eta \leftarrow \gamma\eta$. The scheduler is not reset between stages. In practice the rate was halved twice for VGG-16 (epochs 8 and 14), once for ViT-B/16 (epoch 11) and never for MobileNetV2 and Inception V3 (Fig. 6a).

Progressive fine-tuning. Training runs for three stages of five epochs. In stage s the parameter tensors with index $\geq \tau_s$ (in the framework's parameter order) are trainable, with $\tau = (10, 5, 0)$ for ViT-B/16 and VGG-16, $(249, 120, 0)$ for Inception V3 and $(149, 70, 0)$ for MobileNetV2. This defines nested trainable sets $\boldsymbol{\theta}^{(1)} \subset \boldsymbol{\theta}^{(2)} \subset \boldsymbol{\theta}^{(3)}$, the last being the full network. Gradual unfreezing limits early destruction of pre-trained features [32, 33]. Because the boundaries are tensor indices rather than layers, their effect differs between architectures: stage 1 already trains 96.4% of the ViT-B/16 parameters and 99.1% of VGG-16 (only the embedding and the earliest tensors are frozen), whereas for MobileNetV2 and Inception V3 it restricts training to the last blocks (Table 2). The schedule is therefore genuinely progressive only for the two latter networks.

Table 3 lists the full configuration. It is fixed and shared across models; we do not report a hyper-parameter search.

4.5 Uncertainty quantification

With a 481-image test set, point estimates carry substantial sampling error. We reconstruct the 481 (true, predicted) pairs of each model from its confusion matrix and draw $B = 10{,}000$ bootstrap resamples; the 2.5th and 97.5th percentiles of accuracy and macro-F1 give 95% intervals [34]. Per-class recalls use Wilson intervals [35]. Because the pairs are not aligned across models, the intervals are marginal and cannot replace a paired test (e.g. McNemar's), and they ignore training-run variance.

5. Results

5.1 Overall performance

Table 4 and Fig. 4a summarise the results. ViT-B/16 and Inception V3 both classify 449 of 481 images correctly (93.35%);

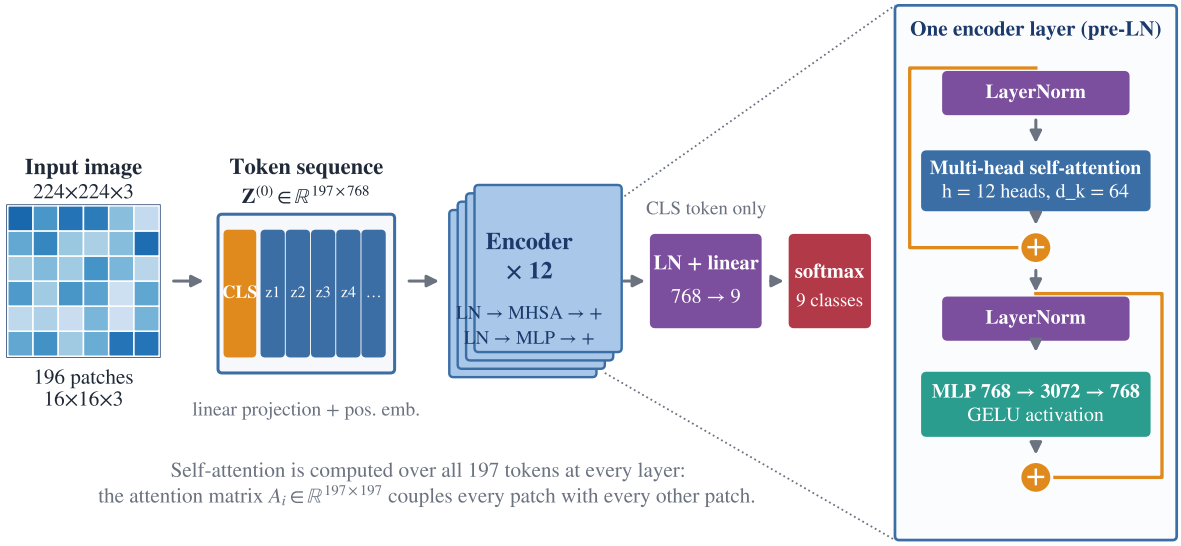


Figure 3. ViT-B/16. The image is cut into 196 patches, linearly projected, prefixed with a learnable CLS token and combined with position embeddings; twelve pre-LN encoder layers follow, and only the CLS output feeds the 9-way head.

Table 3. Training configuration (all models unless stated).

Item	Setting
Optimiser	AdamW, $\eta = 10^{-4}$, $\lambda = 10^{-2}$, default β and ε
Loss	weight-normalised cross-entropy, $w_k = 1/n_k$
Schedule	3 stages × 5 epochs; reduce-on-plateau (max, p=3, γ=0.5)
Batch size	32 (20 for Inception V3)
Precision	fp16 autocast + gradient scaling
Selection	best validation weighted F1 over all 15 epochs
Reproducibility	seed 42 (Python, NumPy, PyTorch, CUDA); single run per model
Gradient clipping	not used

ViT-B/16 has the highest macro-F1 (93.86% versus 93.59%) and macro-precision, but the 0.27-point difference is about eight times smaller than the interval half-width (about 2.2 points), so we cannot rank the two. VGG-16 and MobileNetV2 trail by 2.5 and 4.8 accuracy points; their intervals overlap with the leaders at the edges, so even this gap is only moderately supported. The ranking by validation weighted F1 (ViT-B/16 0.936, Inception V3 0.930, VGG-16 0.912, MobileNetV2 0.907) agrees with the test ranking. The validation-to-test drop in weighted F1 is small for the first three (ViT-B/16 −0.3, VGG-16 −0.3 points; Inception V3 +0.3) and larger for MobileNetV2 (−2.2 points), so checkpoint selection on the validation set did not noticeably overfit it for the leading models.

Cost. ViT-B/16 requires 2.2× the training time and 3.9× the parameters of Inception V3 (3.1× the GFLOPs) for the same accuracy, and 38× the parameters of MobileNetV2 for a 4.8-point accuracy gain. On this dataset the transformer is therefore not on the accuracy–cost Pareto front above the best CNN.

Comparison with the original study. Single et al. report 89.19% for Inception V3 trained on RealWaste [1]. Our Inception V3 reaches 93.35%, but the protocols differ in test images (their separate audit hold-out versus our random hold-out of the public release), resolution (524^2 versus 299^2), augmentation, loss and optimiser, so the 4.2-point difference should not be attributed to any single design choice.

5.2 Per-class behaviour

Fig. 4b and Table 5 give per-class scores. ViT-B/16 attains the best F1 on Cardboard, Glass (tied), Miscellaneous Trash, Paper and Textile Trash; Inception V3 is best on Metal, Plastic and Vegetation, and VGG-16 on Food Organics. Miscellaneous Trash is the weakest class for every model, as in the original study, and ViT-B/16 recovers it best: recall 84.0% (42/50; Wilson interval 71.5–91.7%) versus 78.0% for Inception V3 and VGG-16 and 72.0% for MobileNetV2, with a precision of 95.5% that shows a conservative decision rule (precision above recall, the tendency noted by Single et al.). With only 50 test images per class these differences amount to a few images and are suggestive rather than significant. Paper is the easiest class for ViT-B/16 (F1 97.0%), as it was for the original authors' best model.

5.3 Error structure

The models make 32 (ViT-B/16), 32 (Inception V3), 44 (VGG-16) and 55 (MobileNetV2) errors (Fig. 5). Errors that involve Miscellaneous Trash, as true or as predicted class, account for 10 (31%), 16 (50%), 15 (34%) and 24 (44%) of them, respectively; Metal↔Plastic confusions contribute 9, 5, 9 and 8 errors. Hence one residual class plus one material pair explain roughly 55–65% of all mistakes in every model. Architecture-specific patterns also appear: ViT-B/16 misclassifies Plastic as Metal six

Table 4. Test results on the 481 held-out images. Brackets are 95% bootstrap intervals; training time covers 15 epochs including validation on one GPU.

Model	Accuracy (%)	95% CI	Macro-F1 (%)	95% CI	Macro-P / R (%)	Weighted F1 (%)	Train (min)	s/epoch
MobileNetV2	88.57	[85.7, 91.3]	88.80	[85.8, 91.5]	88.40 / 89.59	88.49	17.3	66–72
VGG-16	90.85	[88.2, 93.4]	91.47	[88.8, 93.8]	91.96 / 91.18	90.83	25.4	81–124
Inception V3	93.35	[91.1, 95.4]	93.59	[91.2, 95.7]	93.98 / 93.30	93.29	31.7	104–162
ViT-B/16	**93.35**	[91.1, 95.4]	**93.86**	[91.5, 95.9]	**94.37** / 93.51	**93.35**	69.5	274–280

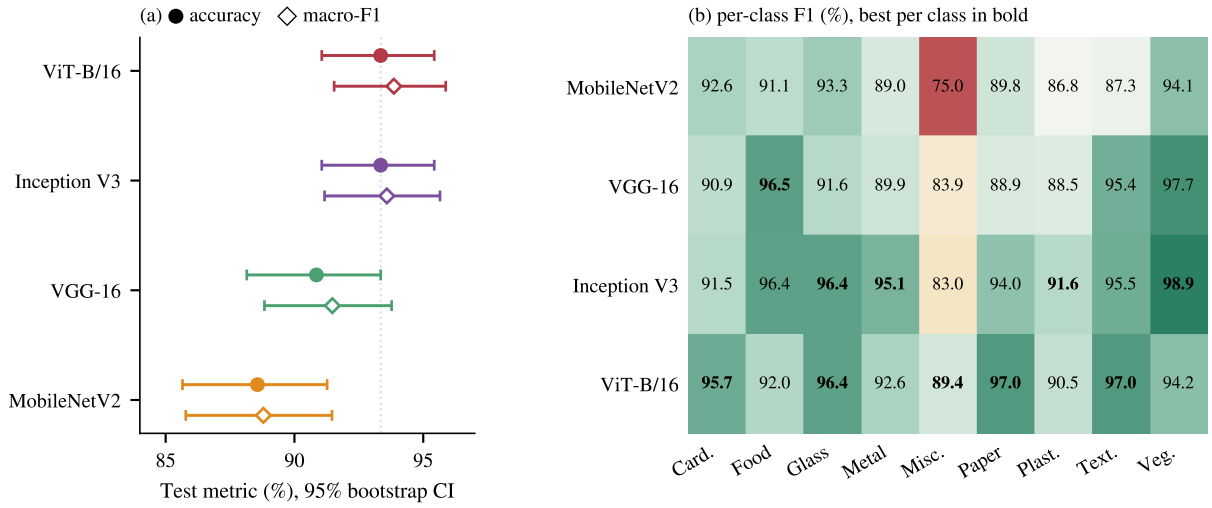


Figure 4. (a) Test accuracy and macro-F1 with 95% bootstrap intervals. (b) Per-class F1; the best model per class is in bold.

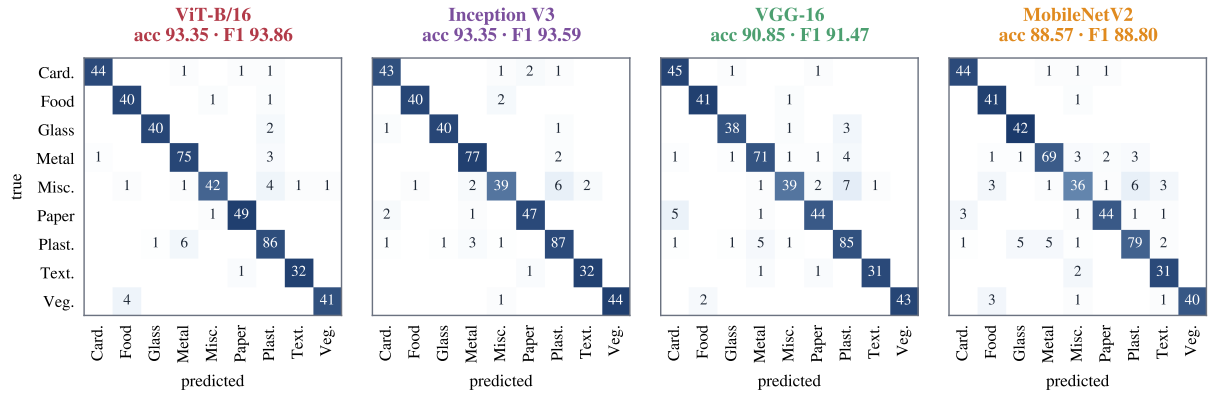


Figure 5. Test confusion matrices (counts; shading is row-normalised recall). The off-diagonal mass concentrates in the same places for all four models: Miscellaneous Trash, Metal↔Plastic, and Vegetation→Food Organics.

times and Vegetation as Food Organics four times (versus none for Inception V3), but has the fewest Miscellaneous Trash errors; VGG-16 confuses Paper with Cardboard five times. These are the same weak points the original authors identified for their Inception V3 [1], which reappear unchanged under a different architecture family.

5.4 Training dynamics and the contribution of each stage

The best validation weighted F1 reached at the end of stages 1, 2 and 3 was 0.790, 0.906 and 0.907 for MobileNetV2; 0.883, 0.905 and 0.912 for VGG-16; 0.908, 0.917 and 0.930 for Inception V3; and 0.925, 0.931 and 0.936 for ViT-B/16 (Fig. 6a). The pattern follows the unfreezing geometry of the previous section. MobileNetV2, whose first stage trains only the last block and the head, gains 11.6 points when stage 2 releases the middle of the network and almost nothing afterwards (0.1); Inception V3 improves steadily (0.9, 1.3); ViT-B/16 and VGG-16, which are almost fully trainable from stage 1, show small increments (0.6/0.5 and 2.2/0.7), and ViT-B/16 is already at 0.925 after five epochs, the highest stage-1 value, which reflects the quality of its pre-trained representation. The training–validation gap at epoch 15 is 5.3, 7.2, 4.4 and 5.5 points for MobileNetV2, VGG-16, Inception V3 and ViT-B/16 (Fig. 6c): all models fit the augmented training set much better than the validation set, and

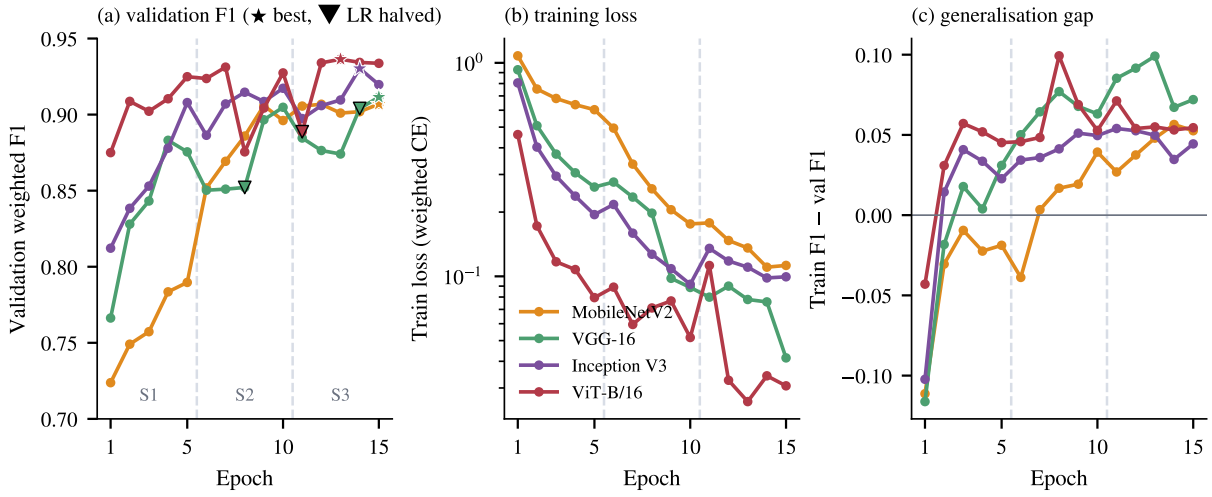


Figure 6. Training dynamics over the 15 epochs (dashed lines: stage boundaries). (a) Validation weighted F1 with the selected checkpoint (⋆) and learning-rate halvings (▽). (b) Training loss. (c) Gap between training and validation weighted F1.

Table 5. Per-class test metrics of ViT-B/16 (%).

Class	Precision	Recall	F1	Support
Cardboard	97.78	93.62	95.65	47
Food Organics	88.89	95.24	91.95	42
Glass	97.56	95.24	96.39	42
Metal	90.36	94.94	92.59	79
Misc. Trash	95.45	84.00	89.36	50
Paper	96.08	98.00	97.03	50
Plastic	88.66	92.47	90.53	93
Textile Trash	96.97	96.97	96.97	33
Vegetation	97.62	91.11	94.25	45
Macro average	94.37	93.51	93.86	481

VGG-16 overfits the most. Validation F1 is noisy (for instance a drop to 0.876 for ViT-B/16 at epoch 8), which is another reason to treat differences below a point as noise.

6. Discussion

What the evidence supports. Under one protocol, a pre-trained ViT-B/16 matches the best CNN in accuracy and is nominally ahead in macro-F1, with the largest advantage on the hardest class. The data do not support the stronger claim that global attention makes ViT superior on landfill imagery: the leading intervals overlap almost completely, results come from a single training run per model, and no experiment isolates the mechanism. The hypothesis remains plausible: global interaction from the first layer [4] is a natural fit for items whose identity depends on context and shape, such as crushed or contaminated objects, while Inception V3's multi-scale branches are equally effective on texture-dominated classes such as Vegetation and Metal. Testing it would require attention-based attribution on the misclassified images and ablations with matched capacity.

The bottleneck is mostly in the labels. Because Miscellaneous Trash and Metal–Plastic dominate the errors of all architectures, and the same confusions appeared in a separate study with different models [1], further backbone changes are unlikely to yield large gains. The more promising directions are data-centric: splitting Miscellaneous Trash into sub-classes, adding a reject/open-set option for items that fit no class, enlarging the Textile Trash and Cardboard classes, and group-aware collection of images so that evaluation reflects new items rather than new views of known ones.

Which pipeline steps mattered. We have no ablations, so we only report what the logs show: (i) split-before-augment is a validity safeguard rather than an accuracy booster, and removing it would inflate any score (Eq. 2); (ii) the gains from progressive unfreezing depend on the tensor-index boundaries, and for ViT-B/16 and VGG-16 the schedule is effectively full fine-tuning from the first stage; (iii) plateau-driven decay acted rarely (three halvings in four runs), so the learning rate was mostly constant at 10^{-4}; (iv) checkpoint selection transferred well to the test set for the three best models. Ablating class weighting, the unfreezing indices and the offline augmentation would quantify these contributions.

Deployment. If throughput matters more than the last two points of accuracy, MobileNetV2 (2.2 M parameters, 0.3 GFLOPs, 88.6%) and Inception V3 (21.8 M, 93.35%) are the economical choices. Our notebook contains an ONNX export routine (opset 18, dynamic batch axis) that we exercised on a DenseNet121 model; we report no ONNX Runtime latencies for the four evaluated models.

Limitations. (1) One random split and one run per model: no variance over seeds or folds. (2) Random splitting without group information (Sec. 3). (3) The validation set contains offline-augmented copies. (4) Bootstrap intervals are marginal, reconstructed from confusion matrices, and ignore training variance. (5) Comparisons with published numbers are indicative only because test sets and protocols differ. (6) DenseNet121 was also trained in our notebook, but its test metrics are not part of the compared set. (7) Parameter and FLOP figures for the

baselines are taken from the framework documentation rather than measured on our hardware.

7. Conclusion

On the RealWaste landfill dataset, a fine-tuned ViT-B/16 reaches 93.35% accuracy and a 93.86% macro-F1, the best macro-F1 among four ImageNet-pre-trained backbones under a leakage-aware protocol, but it ties with Inception V3 in accuracy and the difference between them lies well inside the sampling uncertainty of a 481-image test set. All architectures share the same dominant errors, namely Miscellaneous Trash and the Metal–Plastic pair, which suggests that improving labels and data collection will help more than changing the backbone. Priorities for future work are multi-seed or cross-validated evaluation with group-aware splits, ablations of the pipeline components, attention-based analysis of the misclassified images, and a measured ONNX Runtime benchmark.

Acknowledgements and data availability. This work builds entirely on the RealWaste dataset created by Sam Single, Saeid Iranmanesh and Raad Raad (University of Wollongong), made available at https://github.com/sam-single/realwaste; we thank the authors for releasing it. The images used here are those of the public release.

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